

a inteligência artificial no futuro do ensino superior: desafios e oportunidades

4.º Encontro Aberto do Conselho Geral
da Universidade de Aveiro



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AI-Powered Storytelling Based on Cultural Heritage Archives and Collective Memories

Introduction

Cultural heritage relies on storytelling, traditionally shared in personal interactions. Digital archives now preserve and categorize this knowledge, accessible via curated experiences or search engines [1]. AI-driven platforms, such as Google Arts & Culture, improve engagement but rely on editorial curation. Studies [2,3] highlight the rise of digital storytelling but are scarce in automated generation solutions. Although Artificial Intelligence (AI) storytelling tools are proliferating, the focus is on fictional stories, limiting factual accuracy. Polariscope, a R&D project linking academia and archives, is trying to bridge this gap. It proposes a solution that integrates AI-driven recommendations and generative models to create metadata-based narratives, to enhance engagement with cultural heritage content and collective memories.

Objectives

Polariscope proposes a collaborative storytelling platform by bringing together citizens and institutions to foster rich digital experiences through interactive and narrative visualizations around cultural archives and events (Fig. 1 and 2). The platform offers a tool for users to create stories but also integrates AI to automatically generate stories based on archived content. It assists users in curating narratives while leveraging AI-driven recommendation tools and generative models to automate story creation based on the metadata of uploaded resources. The goal is to disseminate and enrich cultural heritage with user content by providing more engaging and accessible storytelling visualizations.

Methodology

The Polariscope platform aims to present multimodal content (text, images, videos and audio) from digital archives and user uploads, to make it publicly accessible in various visualization formats with a focus on storytelling. Each piece of content is enriched with metadata to provide context, enabling AI-driven story creation. The proposed workflow incorporates AI tools across three key steps (Fig. 3). This architecture aims to enhance storytelling by leveraging AI to connect, contextualize and generate meaningful cultural narratives.

- **Step 1:** *Eden AI* extracts keywords, object detection, and speech-to-text data.
- **Step 2:** *Recombee* recommendation system identifies media semantic correlations for supporting story curation or automated story generation.
- **Step 3:** Users manually curate stories or rely on automatically curated ones through *GPT-3.5* integration based on a set of defined parameters.

User testing assessed AI-generated stories about Portuguese cultural themes using different creative settings (GPT-3.5 temperature parameter), analysing engagement and accuracy of such stories. These stories were presented among others manually curated. The study [4] was carried with 10 postgraduate students in the field of Communication Science Technologies, aged 18-35, selected through a non-probability convenience sampling.

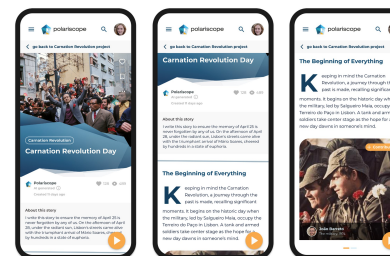


Fig. 1 – Story page navigation (vertical scrollytelling mode).

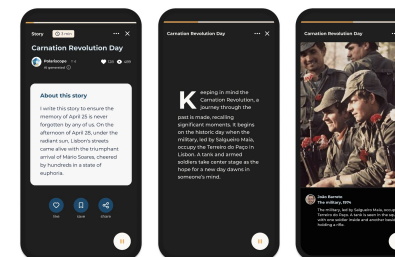


Fig. 2 – Story page navigation (presentation mode).

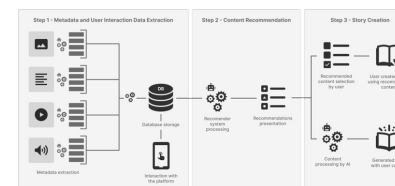


Fig. 3 – System architecture workflow for AI integration.

Results

Participants rated each story on a seven-parameter scale (Interest, Engagement, Quality, Comprehensibility, Fluency, Informativeness and Accuracy) based on Fu et al.'s evaluation framework [5]. Creative outputs (high Temperature) were preferred for stories on local traditions (e.g. São Gonçalinho festivities in Aveiro), while factual topics (e.g., April 25th) benefited from lower Temperature values (less creative stories). Few participants were able to identify the AI-generated texts among non-AI created ones, a result in line with other studies [6] supporting Polariscope's effectiveness in natural-sounding storytelling. Future work includes using adaptive Temperature settings and reinforcement learning-based refinements [7] to optimize AI-assisted storytelling.

Conclusion

Automated storytelling can enhance digital cultural heritage but requires balancing creativity and accuracy. Polariscope integrates AI-driven metadata extraction, recommendation and narrative generation to improve cultural engagement. Future evaluations will be carried in high fidelity implementations with a wider and diverse sample of evaluators.

References

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